

An Introduction to Open Source Programming Languages

Programming is a matter of instructing a computer to perform certain tasks. Code, which is a language that the computer understands, comes in many flavours. The wide choice of coding languages can be pretty overwhelming to a newbie. The author presents an overview of open source programming languages in this article.

Today, programming or coding is not as big a deal as it used to be a few years back. Earlier, this was a method to solve critical problems residing in laboratories or research centres. With advances leading to ease of learning, even a school child can code today, and we have seen considerable enthusiasm from all age groups in learning how to program or code.

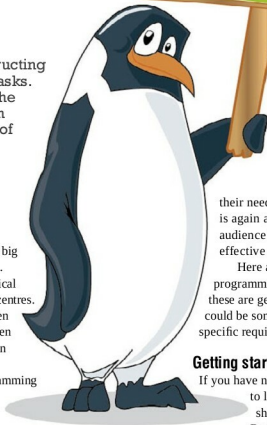
With the emergence of open source programming languages, one is not forced or bound to learn a single language but one can choose from a whole lot of options and can pick something that suits one best. Let me first clarify what the term *open source* actually means. As per the definition given by the Open Source Initiative Certification, "Open source is a computer program or source code available to end users free of cost and they are free to change it accordingly to make it more useful and error-free. Rather than a single proprietary ownership, it will motivate more bug-free and useful code for everyone to use."

A few rules that a programming language must conform to in order to be declared as an open source programming language are listed below:

- Source code must be open and accessible
- Derived works should also remain open source
- Free redistribution
- Integrity of the author's source code should be maintained
- Licence must not restrict other software
- No discrimination against fields of endeavour

How open source works

Open source shows the source code to end users and gives them full freedom to change or modify it as per



their needs. The final edited version is again available for the target audience but will invariably be more effective and error-free by now.

Here are some popular open source programming languages. Of course, these are general suggestions, but there could be some better solutions if your specific requirements are known.

Getting started with HTML and CSS

If you have no coding experience but wish to learn how to code, then you should opt for HTML and CSS.

Rather than a programming language, HTML is a mark-up language. With the help of HTML you will be able to talk to the Web browsers, and you will be embedding text, images and links into your code which a browser can understand and then give you a response. HTML code contains tags which are identifiers of the various things that you have put in your code. A simple snippet of HTML code looks like what follows:

```
<!DOCTYPE html>
My First Heading
<html>
<body>
My first paragraph.
<h1>My First Heading</h1>
<p>My first paragraph. </p>
</body>
</html>
```

On the left we have sample HTML code with the proper syntax and tags, and on the right hand side we have the response returned by the Web browser.